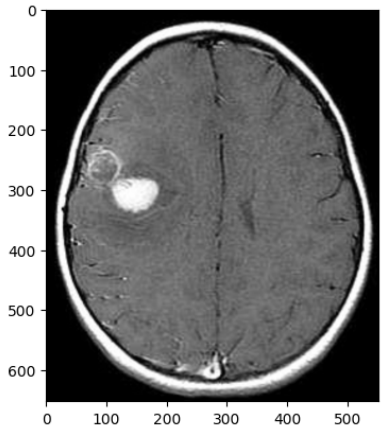
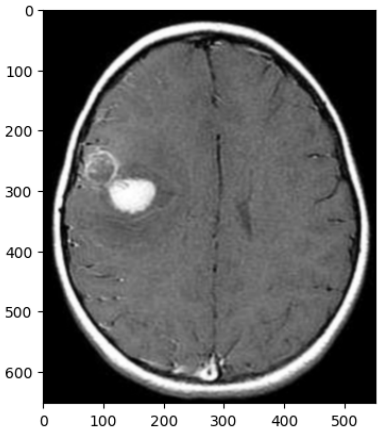
# Analysis Report of Implementation of Segmentation and 4 Basic Morphological Methods for Digital Image Processing

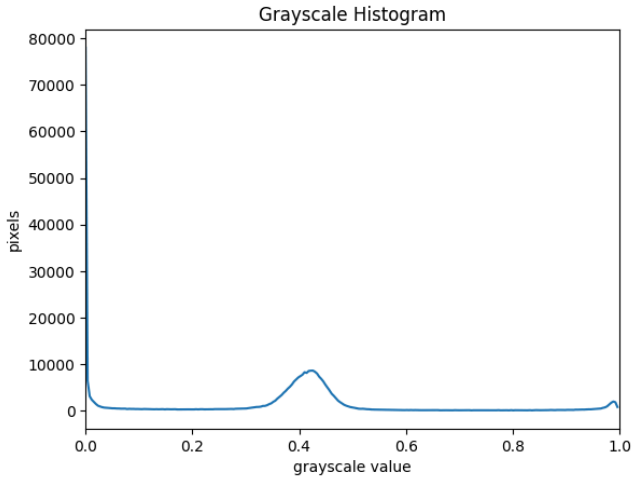
Digital Image Processing Assignment 3

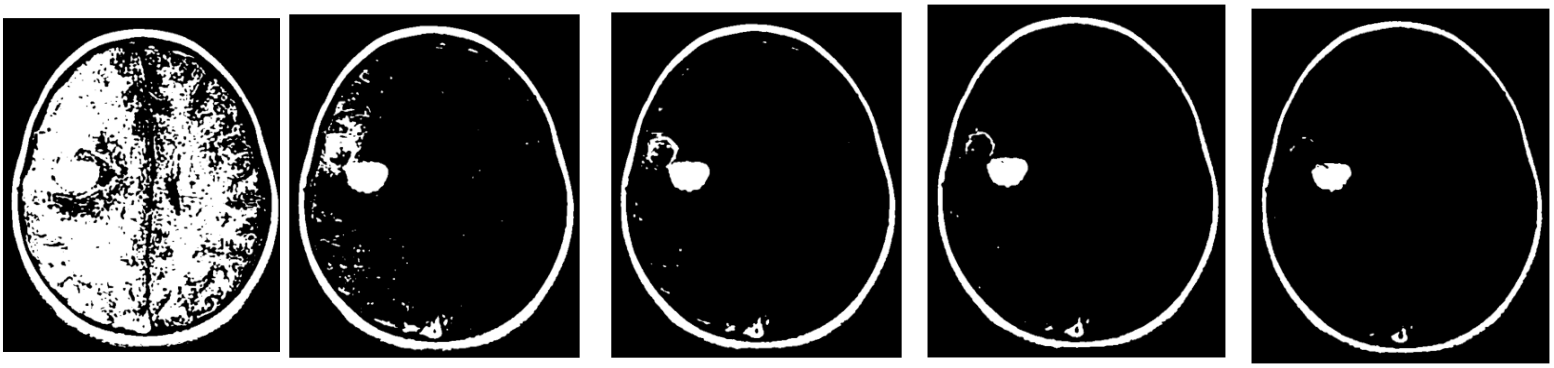
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Part 1. Segmentation



The images above illustrate the steps performed=

The segmentation was performed using the thresholding method. The steps are as follows:

1. The image is turned into a grayscale image:

```
img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
```

This is to make the image have only black and white values for the thresholding process.

2. The Gaussian blur is applied to the image:

```
blurred_shapes = ski.filters.gaussian(img, sigma=1.0)
```

This is to make the intensities more spread and smoothened out in the following histogram as well as remove some noise.

3. A histogram is created of the blurred image:

```
histogram, bin_edges = np.histogram(blurred_shapes, bins=256,
range=(0.0, 1.0))
```

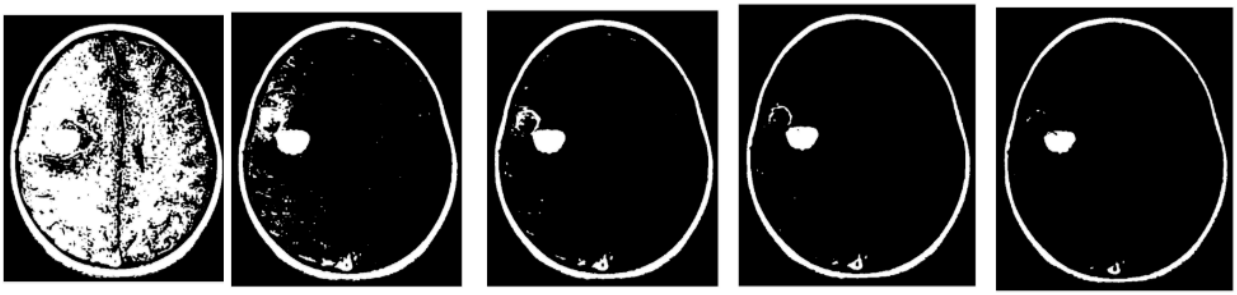
This is to determine at which point the threshold should be determined to segment the tumour. The tumour is very white compared to the rest of the image, so the threshold should be a

brightness intensity value located between 1 (brightest) and the next brightest/intense gray insides (around 0.4)

4. Threshold is determined

```
t = 0.6
binary_mask = blurred_shapes > t
```

Only pixels with a higher brightness value than 0.6 will be converted into value of 1 in the mask, while all other pixels will be converted into a value of 0.

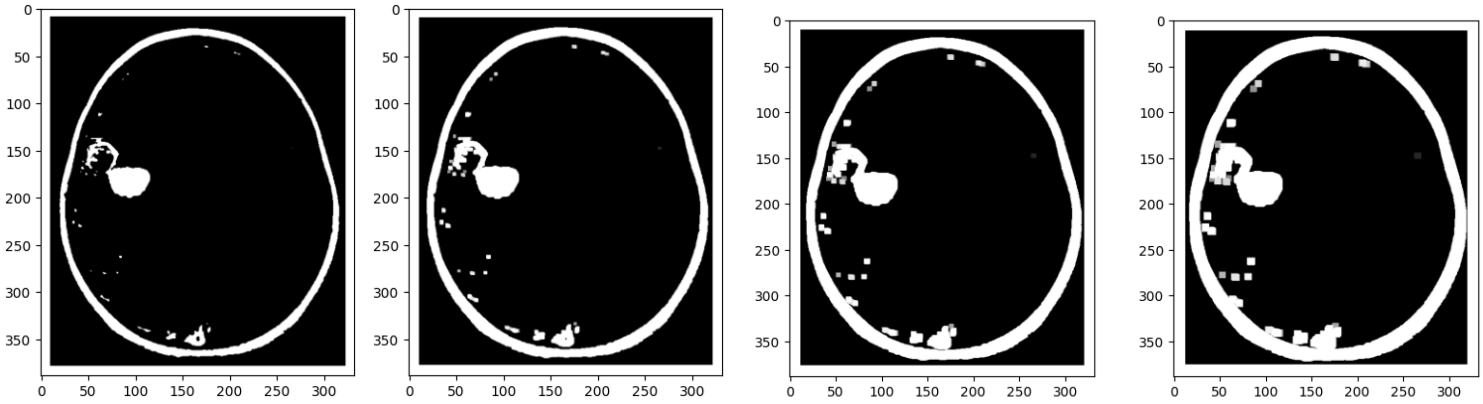


Thresholds of 0.4, 0.5, 0.6, 0.7 and 0.8 respectively.

(The 0.4 threshold is really scary looking…)

Part 2. Morphological Methods

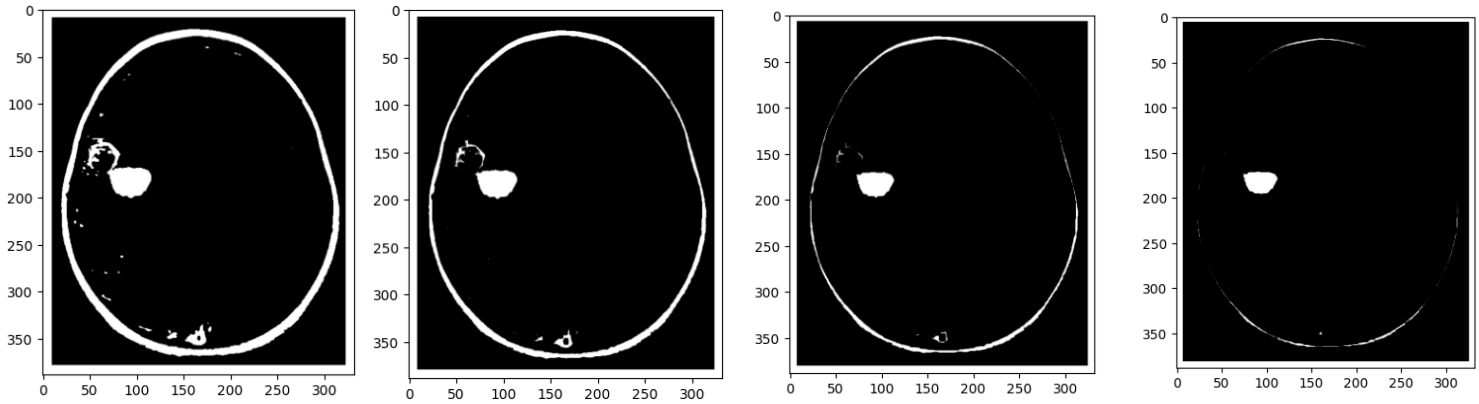
A. Dilation



```
erosion = cv2.dilate(img.copy(), None, iterations=1)
plt.imshow(erosion, cmap='gray')
plt.show()
```

As the iterations grow, the white pixels of the image expand bigger and bigger. The first one is the unmodified image, the second 1 iteration, the third 2 iterations and the fourth is the 3 iterations. Interestingly enough, they look rather squarelike as it expands more. Another interesting feat is that the white border and the circle touch each other in the image that has been iterated three times.

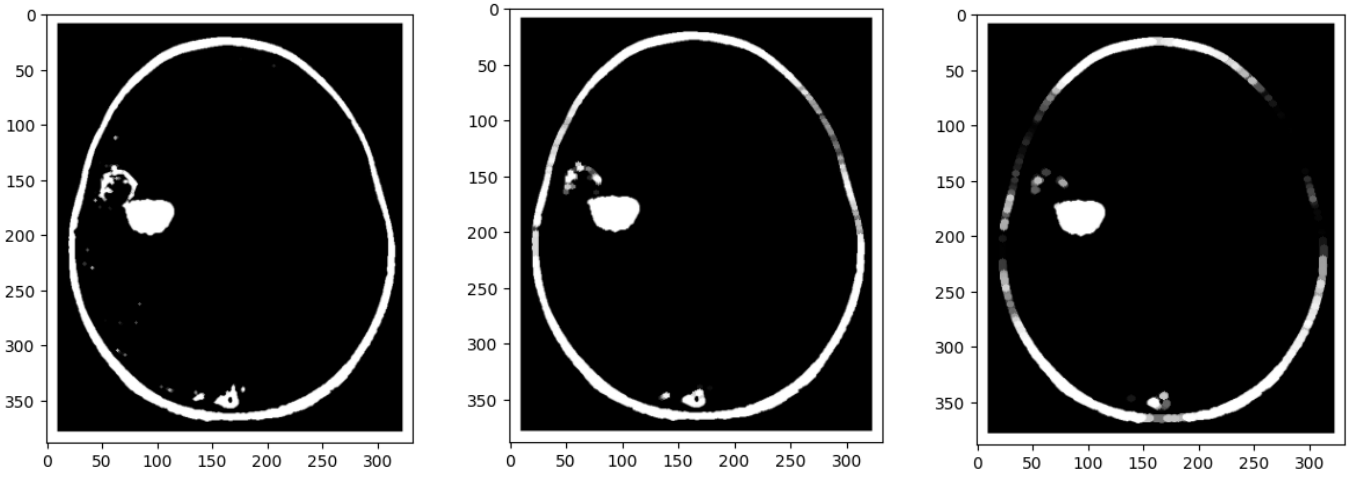
B. Erosion



```
erosion = cv2.erode(img.copy(), None, iterations=1)
plt.imshow(erosion, cmap='gray')
plt.show()
```

The first one is the unmodified image, the second 1 iteration, the third 2 iterations and the fourth is the 3 iterations. The white parts get smaller and smaller until barely anything remains. Most notably, the more “dense” the white part is, the more it survives until the end. Additionally, the tumor is the only thing that remains visible and identifiable after a three times iteration.

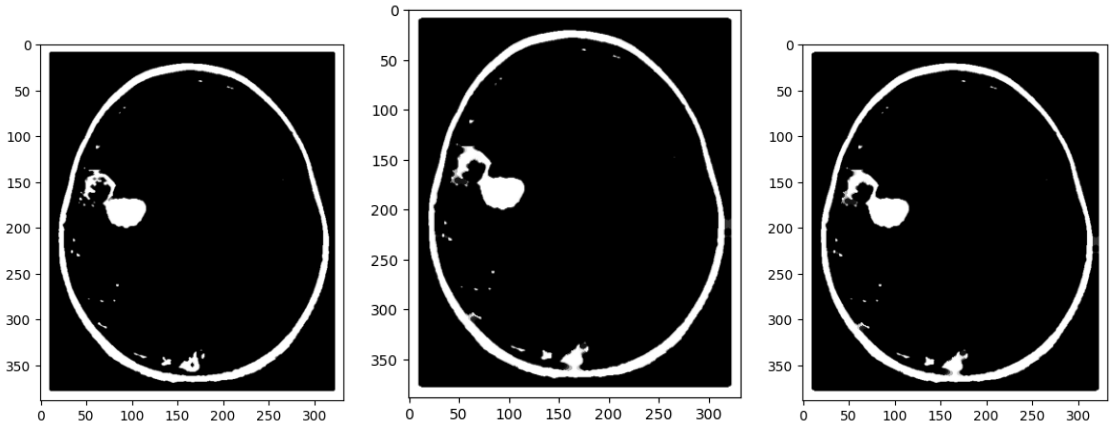
C. Opening



3x3, 5x5, and 7x7 kernel sizes, respectively.

Erosion and then dilation was used on the same image using the same kernel. In this case, an elliptical kernel was used. The effect removes much of the noise but maintains mostly the same shape, but as the kernel size gets bigger, details become lost and the lines and details blur and bleed into each other with minor defects.

D. Closing



3x3, 5x5, and 7x7 kernel sizes, respectively.

Dilation and then erosion was used on the same image using the same kernel. In this case, an elliptical kernel was used. The effect removes small holes but makes objects bleed into each other and lose detail, though on a lesser level than opening.

3. Analysis

It is clear that the erosion or opening is the most suitable method for tumor identification in this scenario as it removes noise most effectively compared to the other two methods; dilation makes the minor spots grow larger and more enhanced in size, which isn't good for identifying the tumor and causes confusion/mistaken tumor identity, while closing doesn't remove the small spots and only smooths out small details from large existing white areas. Erosion reduces the amount of spots significantly but can also remove large features entirely, while opening blurs the features with the background but keeps them, therefore making them both equal.